

COM3031-Probabilistic Machine Learning Workshop: Variational Autoencoders (VAE)

Overview

In this workshop, you will work through a Jupyter notebook to train and test a Variational Autoencoder (VAE) on MNIST using the ELBO objective (reconstruction + KL).

We also include a plain Autoencoder (AE) as a quick baseline. For a VAE, sampling $z \sim \mathcal{N}(0, I)$ and decoding is a normal way to generate new images. For an AE, decoding a randomly sampled z is not reliable, so we only use it to contrast with the VAE.

Questions

Question 1. VAE on MNIST. Use the provided Jupyter notebook for all parts of this question.

- (a) Train the VAE and identify the reconstruction term and the KL term in the code.

$$\mathcal{L}_{\text{VAE}}(x) = \mathcal{L}_{\text{recon}}(x, \hat{x}) + \beta D_{\text{KL}}(q_{\phi}(z | x) \| p(z)).$$

- (b) Visualise reconstructions on held-out examples.
- (c) Generate new samples by drawing $z \sim \mathcal{N}(0, I)$ and decoding.
- (d) Do a latent interpolation between two digits and comment on what changes smoothly.

Question 2. AE baseline. This question uses the AE section of the same notebook.

- (a) Train the AE and compare reconstruction quality to the VAE.
- (b) Compare “generation” behaviour: (i) sample $z \sim \mathcal{N}(0, I)$ then decode, (ii) encode a real image then decode, (iii) (optional) fit a Gaussian to encoded training latents and sample from it.
- (c) Compare AE vs VAE interpolation behaviour qualitatively.

Question 3. Optional quick experiments.

- (a) Change β (e.g. 0.1, 1, 4) and compare reconstructions vs samples.
- (b) Change the latent dimension and compare samples and interpolations.